

Language Modeling

Lecture 2

Language Modeling

- Predict next word

This is the

house, did, rat, then ?

Language Modeling

- A model that assign a probability to each possible next word

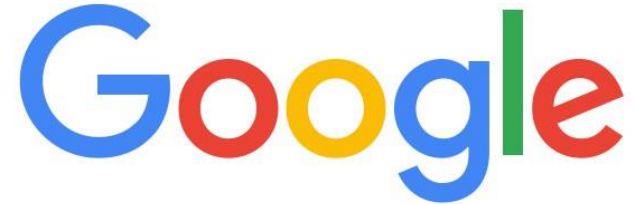
Language Modeling

- A model that assigns a probability to an entire sentence

all of a sudden I notice three guys standing on the sidewalk

on guys all I of notice sidewalk three a sudden standing the

You use Language Models every day!



what is the | 

what is the **weather**
what is the **meaning of life**
what is the **dark web**
what is the **xfi**
what is the **doomsday clock**
what is the **weather today**
what is the **keto diet**
what is the **american dream**
what is the **speed of light**
what is the **bill of rights**

[Google Search](#) [I'm Feeling Lucky](#)

Applications of Language Modeling

- Speech Recognition
 - I saw a van
 - eyes awe of an
- It's not easy to wreck a nice beach.
- It's not easy to recognize speech.
- It's not easy to wreck an ice beach

Applications of Language Modeling

- Machine Translation

他 向 记者 介绍了 主要 内容

He to reporters introduced main content

- he introduced reporters to the main contents of the statement
- he briefed to reporters the main contents of the statement
- **he briefed reporters on the main contents of the statement**

Applications of Language Modeling

- Spelling Correction

Their are two midterms

There is mistyped as their

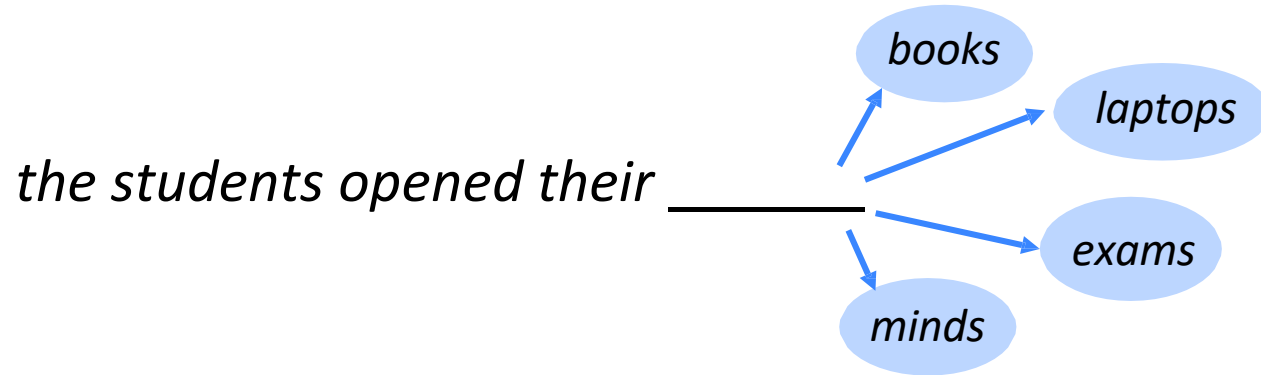
$P(\text{there are}) > P(\text{their are})$

Applications of Language Modeling

- Many more applications
 - Question Answering
 - Summarization etc.

Language Modeling

- **Language Modeling** is the task of predicting what word comes next.



- More formally: given a sequence of words $x^{(1)}, x^{(2)}, \dots, x^{(t)}$, compute the probability distribution of the next word $x^{(t+1)}$:

$$P(x^{(t+1)} \mid x^{(t)}, \dots, x^{(1)})$$

where $x^{(t+1)}$ can be any word in the vocabulary $V = \{w_1, \dots, w_{|V|}\}$

- A system that does this is called a **Language Model**.

Probabilistic Language Modeling

- Goal: compute the probability of a sentence or sequence of words:

$$P(W) = P(w_1, w_2, w_3, w_4, w_5 \dots w_n)$$

- Related task: probability of an upcoming word:

$$P(w_5 | w_1, w_2, w_3, w_4)$$

- A model that computes either of these:

$P(W)$ or $P(w_n | w_1, w_2 \dots w_{n-1})$ is called a **language model**.

How to compute $P(W)$

- How to compute this joint probability:
 - $P(\text{its, water, is, so, transparent, that})$
- Intuition: let's rely on the Chain Rule of Probability

Reminder: The Chain Rule

- Recall the definition of conditional probabilities

$$P(A,B) = P(A) P(B|A)$$

- More variables:

$$P(A,B,C,D) = P(A) P(B|A) P(C|A,B) P(D|A,B,C)$$

The Chain Rule applied to compute joint probability of words in sentence

The Chain Rule in General

$$P(x_1, x_2, x_3, \dots, x_n) = P(x_1)P(x_2 | x_1)P(x_3 | x_1, x_2) \dots P(x_n | x_1, \dots, x_{n-1})$$

$$P(w_1 w_2 \dots w_n) = \prod_i P(w_i | w_1 w_2 \dots w_{i-1})$$

$P(\text{"its water is so transparent"}) =$

$P(\text{its}) \times P(\text{water} | \text{its}) \times P(\text{is} | \text{its water})$

$\times P(\text{so} | \text{its water is}) \times P(\text{transparent} | \text{its water is so})$

How to estimate these probabilities

- Could we just count and divide?

$$P(\text{the} \mid \text{its water is so transparent that}) = \frac{\text{Count}(\text{its water is so transparent that the})}{\text{Count}(\text{its water is so transparent that})}$$

- No! Too many possible sentences!
- We'll never see enough data for estimating these

Markov Assumption



Andrei Markov

- Simplifying assumption:

$P(\text{the } | \text{ its water is so transparent that}) \approx P(\text{the } | \text{ that})$

- Or maybe

$P(\text{the } | \text{ its water is so transparent that}) \approx P(\text{the } | \text{ transparent that})$

Markov Assumption

$$P(w_i \mid w_1 w_2 \dots w_{i-1}) \sim P(w_i \mid w_{i-k} \dots w_{i-1})$$

Simplest case: Unigram model

$$P(w_i | w_1 w_2 \dots w_{i-1}) \sim \prod_i P(w_i)$$

Some automatically generated sentences from a unigram model

fifth, an, of, futures, the, an, incorporated, a,
a, the, inflation, most, dollars, quarter, in, is,
mass

thrift, did, eighty, said, hard, 'm, july, bullish

that, or, limited, the

Bigram model

- Condition on the previous word:

$$P(w_i | w_1 w_2 \dots w_{i-1}) \sim \prod_i P(w_i | w_{i-1})$$

texaco, rose, one, in, this, issue, is, pursuing, growth, in,
a, boiler, house, said, mr., gurria, mexico, 's, motion,
control, proposal, without, permission, from, five, hundred,
fifty, five, yen

outside, new, car, parking, lot, of, the, agreement, reached

this, would, be, a, record, november

N-gram models

- We can extend to trigrams, 4-grams, 5-grams
- In general this is an insufficient model of language
 - because language has **long-distance dependencies**:
“The computer which I had just put into the machine room on the fifth floor crashed.”
- But we can often get away with N-gram models

Estimating bigram probabilities

- The Maximum Likelihood Estimate

$$P(w_i \mid w_{i-1}) = \frac{\textit{count}(w_{i-1}, w_i)}{\textit{count}(w_{i-1})}$$

An example

$$P(w_i \mid w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

<s> I am Sam </s>

<s> Sam I am </s>

<s> I do not like green eggs and ham </s>

$$P(\text{I} \mid \text{<s>}) = \frac{2}{3} = .67$$

$$P(\text{Sam} \mid \text{<s>}) = \frac{1}{3} = .33$$

$$P(\text{am} \mid \text{I}) = \frac{2}{3} = .67$$

$$P(\text{</s>} \mid \text{Sam}) = \frac{1}{2} = 0.5$$

$$P(\text{Sam} \mid \text{am}) = \frac{1}{2} = .5$$

$$P(\text{do} \mid \text{I}) = \frac{1}{3} = .33$$

More examples: Berkeley Restaurant Project sentences

- can you tell me about any good cantonese restaurants close by
- mid priced thai food is what i'm looking for
- tell me about chez panisse
- can you give me a listing of the kinds of food that are available
- i'm looking for a good place to eat breakfast
- when is caffe venezia open during the day

Raw bigram counts

- Out of 9222 sentences

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

Raw bigram probabilities

- Normalize by unigrams:

- Result:

i	want	to	eat	chinese	food	lunch	spend
2533	927	2417	746	158	1093	341	278

	i	want	to	eat	chinese	food	lunch	spend
i	0.002	0.33	0	0.0036	0	0	0	0.00079
want	0.0022	0	0.66	0.0011	0.0065	0.0065	0.0054	0.0011
to	0.00083	0	0.0017	0.28	0.00083	0	0.0025	0.087
eat	0	0	0.0027	0	0.021	0.0027	0.056	0
chinese	0.0063	0	0	0	0	0.52	0.0063	0
food	0.014	0	0.014	0	0.00092	0.0037	0	0
lunch	0.0059	0	0	0	0	0.0029	0	0
spend	0.0036	0	0.0036	0	0	0	0	0

Bigram estimates of sentence probabilities

$$\begin{aligned} P(<s> \text{ I want english food } </s>) = & \\ P(\text{I} | <s>) & \\ \times P(\text{want} | \text{I}) & \\ \times P(\text{english} | \text{want}) & \\ \times P(\text{food} | \text{english}) & \\ \times P(</s> | \text{food}) & \\ = .000031 & \end{aligned}$$

Google N-Gram Release, August 2006

AUG

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All Our N-gram are Belong to You

Posted by Alex Franz and Thorsten Brants, Google Machine Translation Team

Here at Google Research we have been using word [n-gram models](#) for a variety of R&D projects,

...

That's why we decided to share this enormous dataset with everyone. We processed 1,024,908,267,229 words of running text and are publishing the counts for all 1,176,470,663 five-word sequences that appear at least 40 times. There are 13,588,391 unique words, after discarding words that appear less than 200 times.

Google N-Gram Release

- serve as the incoming 92
- serve as the incubator 99
- serve as the independent 794
- serve as the index 223
- serve as the indication 72
- serve as the indicator 120
- serve as the indicators 45
- serve as the indispensable 111
- serve as the indispensable 40
- serve as the individual 234

<http://googleresearch.blogspot.com/2006/08/all-our-n-gram-are-belong-to-you.html>

Google Book N-grams

- <http://ngrams.googlelabs.com/>

Slide Credits

- Lecture Notes, Natural Language Processing by Christopher Manning and Daniel Jurafsky, Stanford University